IP address & Subnets

Subnet mask :- This tells which part is the network portion and which part is host.

ex:- 192.168.1.25

Subnet mask → 255.255.255.0 → This is the host part which can change from 0-255, total of 256.

↓

There 3 no.s tells us the first 3 octets of this ip address is the network portion.

in other words those numbers will not change.

→ A device in this network (192.168.1) will have a total of 256 hosts which means in this network a total of 256 ip addresses can be assigned, so every device in this network gets an unique ip & the unique part will be the host part, so ex:- a device will have 192.168.1.5 as its ip address the other device will have 192.168.1.6 as its ip address in that network.

→ The four sections of ip address are called octets. Each octet = 8 bits

so 192.168.1.25
↓ ↓ ↓ ↓
8 bits 8 bits 8 bits 8 bits

8+8+8+8 = 32 bits

Now Sometimes or most of the times an ip will be written as:-

192.168.1.25/24 → This is called CIDR Notation or prefix length. another way of writing the subnet mask.

/24 → represents the network part in total no. of bits, so 8+8+8 = 24 which means in this ip the first 3 octets are network part

If its written as /16 → means 8+8, so the first two octets represents network part & the last 2 octets can change (host part)

Depending on that we can determine which is the network portion & which is the host portion in an ip address.

ex:- /16 means → 255.255.0.0
/24 means → 255.255.255.0

The IPV4 consist of 4billion ip addresses, at ceretain point we ran out of the ip addresses which is when NAT & private ip addresses were developed.

Instead of every device getting a public ip, each device in a network will receive private ips and the router on that network will receive one public ip assigned by the ISP. Now ~~at~~ devices in ~~other~~ networks can have same private ips and only the router will have a public ip.

If device wants to accus a website it sends request, the router receives & translates the device's private ip to its public ip as source ip so other routers can route the packet when response as private ips ~~are~~ ~~not directly accessible by~~ cannot be reached directly from the internet.

This process of conversion of private → public or otherway is called Network Address Translation (NAT)

Even with that we still ran out of ip addresses which is when we discovered ipV6 (total of 340 undecillion addresses

Most of the devices if using ipv4 have private ip addresses (check in cmd) But for devices with ipv6 addresses they are mostly public ipv6 addresses.

public ips :- Reachable across the internet & globally unique

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Computers don't understand decimal ips like

192.168.1.21 → decimal

It understands binary, ex:-

11000000.10101000.00000001.00010101

↳ binary

How to convert ?

↓

128	64	32	16	8	4	2	1
1	1	0	0	0	0	0	0

pick number 192

128 fits in so mark that 1

128+64 = 192 mark as 1

Mark the remaining 0 as we got the no. 192

so the binary of 192 → 11000000

lets do the 2nd Octet 168

↓

128	64	32	16	8	4	2	1	(2^0 to 2^8 btw)
1	0	1	0	1	0	0	0	↵

128 fits in mark 1

128 + 64 = 192 which is > 168 so doesn't count mark as 0

128 + 32 = 160 < 168 so counts mark as 1

160 + 16 = 176 > 168 so doesn't count

160 + 8 = 168 == 168 counts so mark as 1, we got the number so remaining mark as 0

binary version of 168 → 10101000

we do the same for subnet mask

Subnet :- It means division within or of a network or dividing larger network into smaller networks

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Note :- We have total of 256 in a octet but from that 2 are reserved which leaves us 254 usable ip addresses

Those two reserved for :-
- network address (.0) ← last octet
- Broadcast address (.255)

Subnetting :- chaging the subnet mask value ~~to manipulate with the no. of ip addresses~~ to eventually set more networks from a larger network.

192.168.32.5 → ip
255.255.255.0 → subnet mask

CIDR → 192.168.32.5/24

↓ decimal conversion

11000000.1010 1000.001 00000.00000101
11111111.1111 1111.111 11111.00000000

Here in subnet mask the 0's represent host bits

no. of bits = 8, $2^8 = 256$
= 254 (reserved addresses)
usable ips

Now lets say u need more ip addresses, in that case we manipulate the subnet mask value

~~1100~~
11111111.11111111.11111110.00000000

Notice there's no extra 0. ~~So~~ And as we know '0' represent host bits. Now total no. of 0s = 9

$2^9 = 512$
= 510 usable ips

If need more ips just change subnet mask.

Also the decimal version will now look like

↓
255.255.254.0 and CIDR → 192.168.32.5/23

The previous example was for simplified understanding, but subnetting means getting more networks not ips.

So instead of /24 → /23

In subnetting → /24 → /25 → means now there's more networks

u can still increase the no. of ips, & its just that example will not be considered as subnetting.

Increasing host bits → supernetting
Increasing network bits → subnetting

Example:- lets say u have

192.168.1.0/24

u change it to

192.168.1.0/26 → this means the network bits increased which means u stole 2 bits from host part

1 bit can either be 0 or 1, so $2^1 = 2$

So for 2 bits = $2^2 = 4$

The combinations for those 2 borrowed bits

↓

00
01
10
11

→ 4 possibilities

So that means we have 4 networks

and the new subnet mask:-

11111111.11111111.11111111.11000000

convert that to decimal

↓

255.255.255.(192)

↓

256 - 192 = (64)

no. of host bits

↑

or $2^{(6)} = 64$

↓

This means every new sub-network (subnet) will start after 64 addresses

So → 0 - 63 (we count 0 as well)
64 - 127
128 - 191
192 - 255

So now the subnets are :-

192.168.1.0/26
192.168.1.64/26
192.168.1.128/26
192.168.1.192/26

we got more networks but less hosts bits which is 6, so $2^6 = 64$ which as we saw in example, there are 64 addresses per subnet.

4 networks x 64 = 256 (so its same just divided)

lets say in a situation u got 40 hosts per 3 coffee shops, and now u need to make subnets based on hosts.

↓

Original network → 192.168.1.0/24

to find how many host bits are needed

↓

$2^5 = 32$ addresses (too ~~t~~ not enough)

$2^6 = 64$ addresses (enough) → usable 62

So we need 6 host bits

ipv4 has 32 bits in total = 32-6
= 26

Therefore subnet mask = /26

/26 - /24 = 2 so $2^2 = 4$ subnets

192.168.1.0/26 gives us same four subnets

192.168.1.0/26
192.168.1.64/26
192.168.1.128/26
] - we can use these 3 subnets

192.168.1.192/26 → leave it unused

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u got 172.16.0.0/20

we need 6 branches with each needing 100 ips

answer

↓

First know the third octet can change

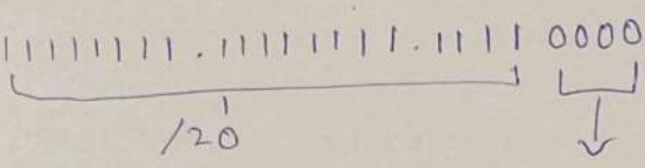
172.16.0.0/20

subnet = 20

32 - 20

= 12

or



$2^4 = 16$

So the third octet will range from 0-15

~~Soon~~ Now calculation

↓

$2^7 = 128$ ips (enough)

32 - 7 = 25

new subnet mask = /25

~~old~~ subtract with old = /25 - /20

= 5

$2^5 = 32$ subnets

we got 32 subnets each having 128 ips

Now remember the third octet originally had 0-15 range *, we always consider the original subnet mask to list out subnets which was /20.

So now, as we got 32 subnets with 128 ips each.

originaly we had ~~32~~ 12 host bits, $2^{12} = 4096$ addresses

now $32 \times 128 = 4096$ so it matches up.

↓

172.16.0.0 → 172.16.0.127

172.16.0.128 → 172.16.0.255

~~we used all for 3rd octet~~

each subnet can contain 128 ips so here we ran out. which means the next starts on

~~172.16.0.~~

172.16.1.0 → 172.16.1.127

↓ continues till

172.16.15.255

which means we had 16 separations 2 subnet each in one separation

$2 \times 16 = 32$ subnets

[Note:- Always consider the original subnet mask when listing out subnets.]

Which concludes:-

To figure out the overall range subnets we refer the original subnet mask and to figure out the range of each subnets we refer the new subnet mask. So we need both.

All the subnets will have new subnet mask to determine this subnet ends in this range thats why after subnetting we write the new subnet mask & not the old one.

Reverse Subnetting—

156.42.19.5/22

find network & broadcast address for this subnet

11111111.11111111.111111**00**.00000000

↳

$2^2 = 4$

Range will lie in

0-3, 4-7, 8-11, 12-15, 16-19, 20-23

The third octet value 19 falls in (16-19) range so pick the 1st for network & last for broadcast and for the fourth octet as all was bits 0 we pick 0 as network & 255 as broadcast.

↓

156.42.16.0 → network address

156.42.19.255 → broadcast address

Subnetting a subnet

lets say u got a network 172.21.42.0/24

↓

u need 4 more networks with requirements

10 hosts
57 hosts
26 hosts
117 hosts

here we will be using VLSM

always choose the largest one first

no hosts	minimum address	subnet
117	128	/25
57	64	/26
26	32	/27
10	16	/28

172.21.42.0/25

↳ This makes two subnets as /24 → /25 = 1
= $2^1 = 2$

172.21.42.0/25 (0-127) ← This one allocated for 117 hosts

172.21.42.128/25 (128-255)

↳ Now we further subnet this second one for other hosts requirements.

57 hosts

↓

172.21.42.128/26 → range

172.21.42.128 – 172.21.42.191 (64) → used for 57 hosts

172.21.42.192 – 172.21.42.255 (64)

↳ This second one will be further subdivided for the remaining hosts.

at the end

↓

26 hosts

↓

172.21.42.192/27

10 hosts

↓

~~172.41~~

172.21.42.224/28